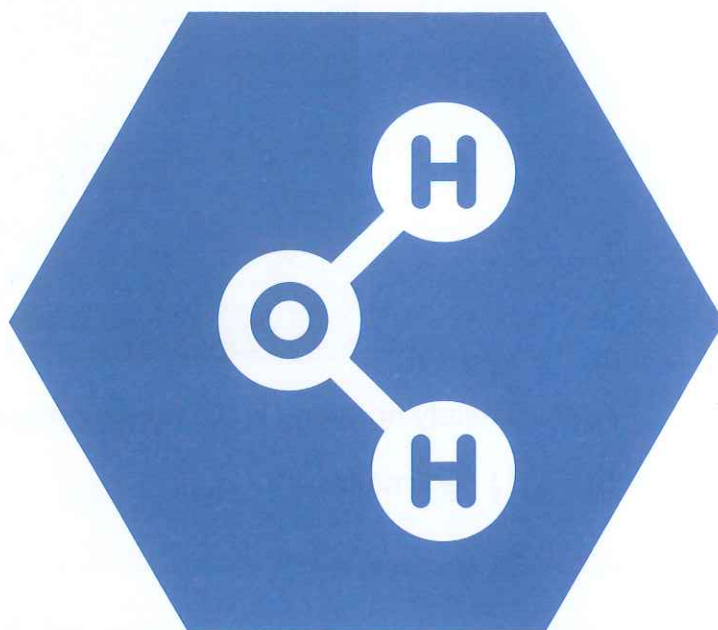




UNSW Global
AUSTRALIA

PAPER A



2015 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

Educational Assessment Australia
eaa.unsw.edu.au

1. This photograph shows a life-size hummingbird. It has a red beak.



Approximately how long is the hummingbird's beak?

- (A) $1\frac{1}{2}$ centimetres (B) 2 centimetres
(C) $2\frac{1}{2}$ centimetres (D) 3 centimetres

2. These photographs show the feet of four different birds.



1



2



3

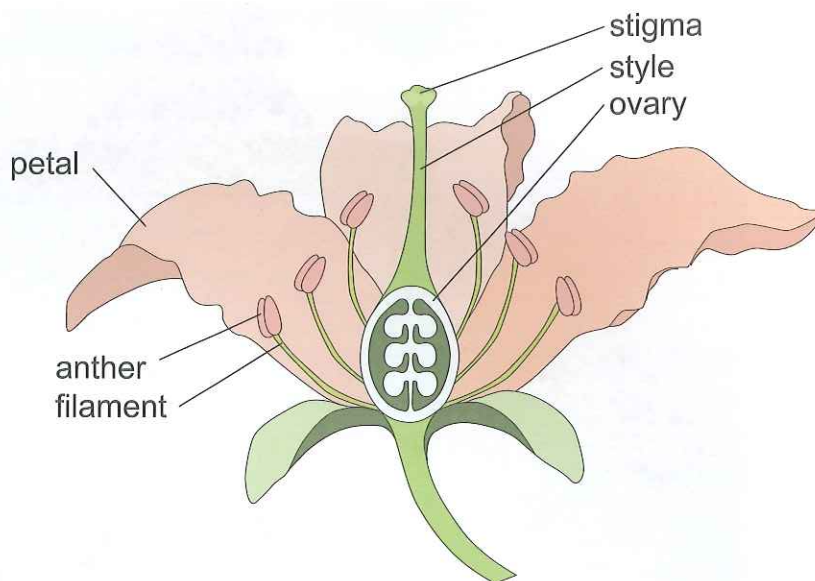


4

Which photographs show bird's feet that are best suited to swimming?

- (A) 1 and 2 (B) 2 and 4 (C) 3 and 4 (D) 1 and 4

3. This diagram shows some parts of a flower.



On top of which part of the flower is the anther found?

- (A) stigma (B) filament (C) style (D) ovary

4. Plants that have flowers can be classified into two groups called monocots and dicots.

Monocot plants have flowers with three or six petals. Dicot plants have flowers with four or five petals.

These photographs show four flowers.



1



2



3

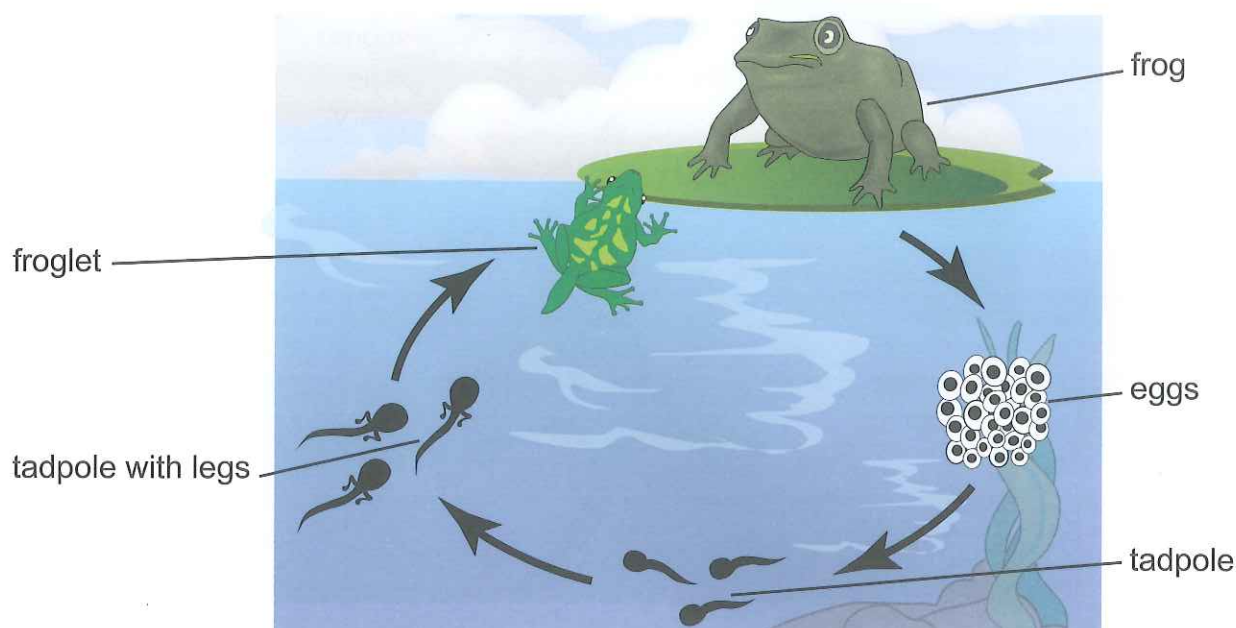


4

Which flowers are from dicot plants?

- (A) 3 and 4 (B) 2 and 3 (C) 1 and 2 (D) 1 and 3

5. This diagram shows the life cycle of a frog.



Based on the diagram, which of the following statements is true?

- (A) Tadpoles are born with legs.
 - (B) Frogs can survive out of water.
 - (C) Frog eggs are attached to plants on dry land.
 - (D) Frogs remain the same colour throughout their life cycle.
6. Jacqui made a cup of tea. She did not add any milk or sugar. Tony squeezed lemon juice into Jacqui's tea. Jacqui saw that the tea turned yellow. Tony thinks the tea will taste sour because of the lemon juice.

Which of the following is an observation?

- (A) Tony wants to taste Jacqui's tea.
- (B) Jacqui likes lemon in her tea.
- (C) The tea will taste sour.
- (D) The tea turned yellow.

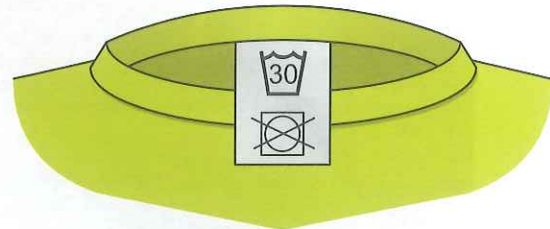
7. Most pieces of clothing have a label with symbols that show how to clean it.

This chart shows the meaning of some of these symbols.

Fabric Care Symbols

MACHINE WASH	TUMBLE DRY	DRY	IRON	DRY CLEAN
TEMPERATURE	HEAT SETTING		TEMPERATURE (Dry or Steam)	
30 Cool/Cold	No Heat	Line Dry/ Hang to Dry	Low	Dry Clean
40 Warm	Low	Drip Dry	Medium	Do Not Dry Clean
60 Hot	Medium	Dry Flat	High	
CYCLE	High	Do Not Wring	OTHER	
Normal	Any Heat		No Steam	
Permanent Press	OTHER		Do Not Iron	
Delicate/Gentle	Do Not Tumble Dry			

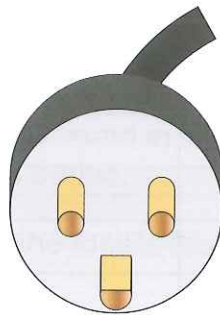
The diagram shows the label attached to Sean's shirt.



Using these instructions, how should Sean clean his shirt?

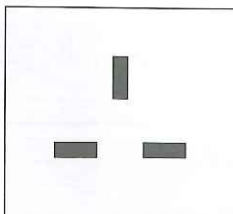
- (A) wash in cool water; do not dry clean
- (B) wash in warm water; do not dry clean
- (C) wash in cool water; do not tumble dry
- (D) wash in warm water; do not tumble dry

8. Many electrical appliances have to be plugged into a power outlet. This diagram shows the plug of an electrical appliance.

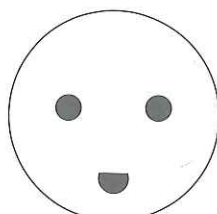


Into which power outlet will this plug fit?

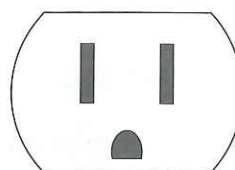
(A)



(B)



(C)



(D)



9. Some students wanted to investigate what causes day and night on Earth. They used a lamp to represent the Sun. They shone the light from the lamp onto a globe as they rotated the globe.



Which statement is supported by their investigation?

- (A) Night occurs on the side of Earth that is facing the Sun.
- (B) Night occurs all over Earth's surface at the same time.
- (C) Light from the Sun does not cover all of Earth's surface at the same time.
- (D) Day occurs on both sides of Earth's surface at the same time.

For questions 10 and 11 use the information below.

The table shows the fastest speed that some animals can run.

Animal	Fastest speed (km/h)
cheetah	120
emu	48
giraffe	51
greyhound dog	63
human	44
polar bear	30

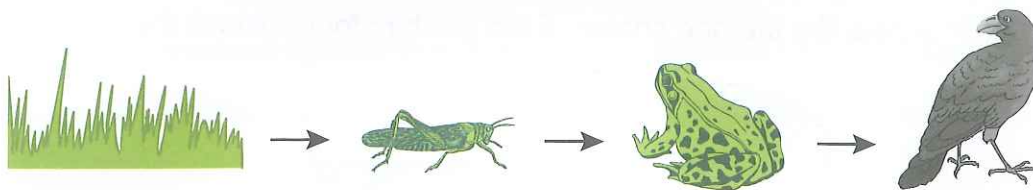
10. Which animals can run faster than a giraffe?

- (A) emu and cheetah
- (B) greyhound dog and human
- (C) cheetah and greyhound dog
- (D) human and emu

11. How much faster than a human can a cheetah run?

- (A) 120 km/h
- (B) 86 km/h
- (C) 76 km/h
- (D) 44 km/h

12. This diagram shows a food chain.



The food chain shows that:

- grasshoppers eat plants
- frogs eat grasshoppers
- crows eat frogs

If all the plants were removed, which living thing will go hungry first?

- (A) plants (B) grasshoppers (C) frogs (D) crows

13. Lily has three rocks – rock 1, rock 2 and rock 3.

She wants to see which rock is the hardest. She rubs rock 1 and rock 2 together. Rock 2 now has a scratch on it.

She rubs rock 2 and rock 3 together. Rock 3 now has a scratch on it.

Which rock is the hardest?

- (A) rock 1 (B) rock 2
(C) rock 3 (D) All rocks have the same hardness.

14. Fossils are the remains of living things found in layers of rock. The older the fossil, the deeper the layer of rock in which it is buried.

This is a diagram of rock layers and the fossils found in them.



Which of these is the oldest fossil?

(A)



(B)



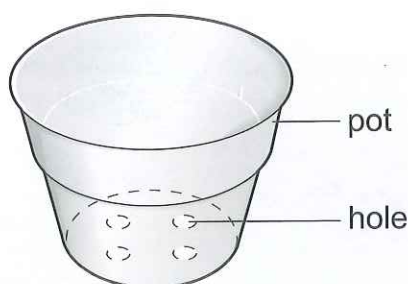
(C)



(D)



15. Jane wanted to find out which type of soil holds the largest amount of water. She used four identical pots like the one shown. Each pot had four holes in the bottom.



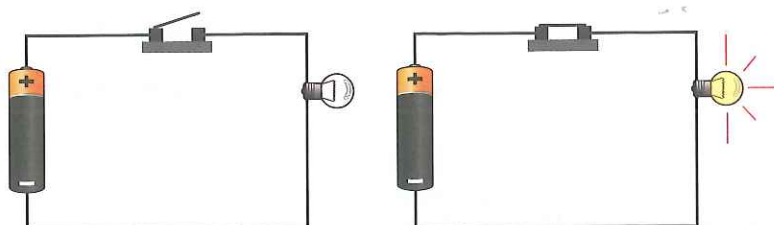
Jane filled each pot with an equal amount of soil. Each pot had a different type of soil. Then she added the same amount of water to each pot. Some water leaked out of each pot.

How can Jane find out which type of soil holds the largest amount of water?

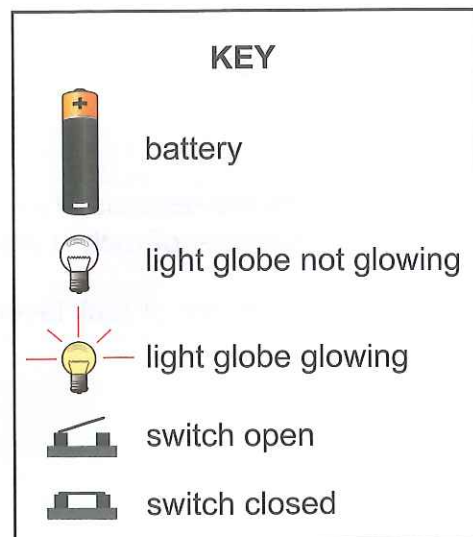
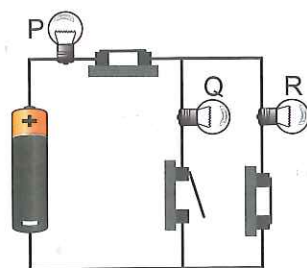
- (A) measure how much water comes out of each pot
- (B) measure the amount of soil falling from each pot
- (C) test which soil loses water first
- (D) test which soil feels wet

16. Sally is investigating electrical circuits.

She uses some diagrams to help her work out what is needed to make a light globe glow.



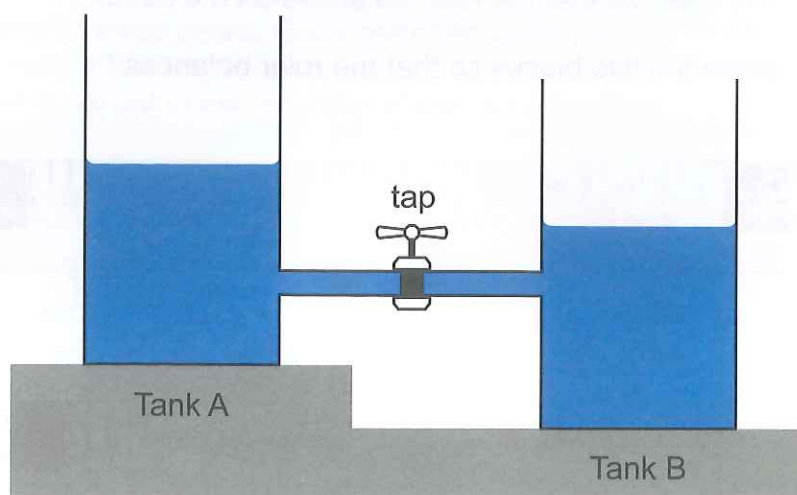
Sally wires up this circuit. The light globes all work and the battery produces enough electricity to make them all glow.



Which light globes in her circuit will glow?

- (A) P and R only
- (B) P and Q only
- (C) Q and R only
- (D) P, Q and R

17. This diagram shows two tanks connected by a pipe with a tap that controls the flow of water.

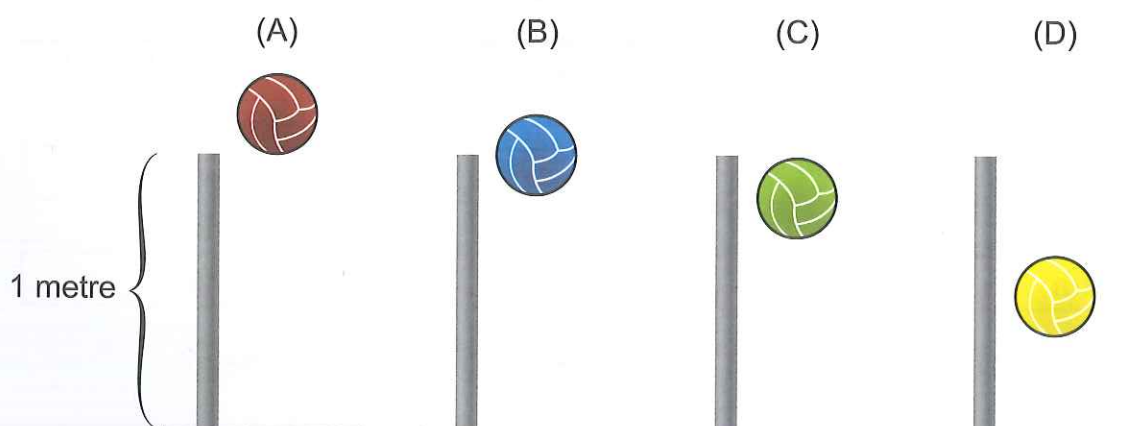


When the tap is opened, what will happen to the water level in each tank?

	Tank A	Tank B
(A)	↑	↑
(B)	↑	↓
(C)	↓	↑
(D)	↓	↓

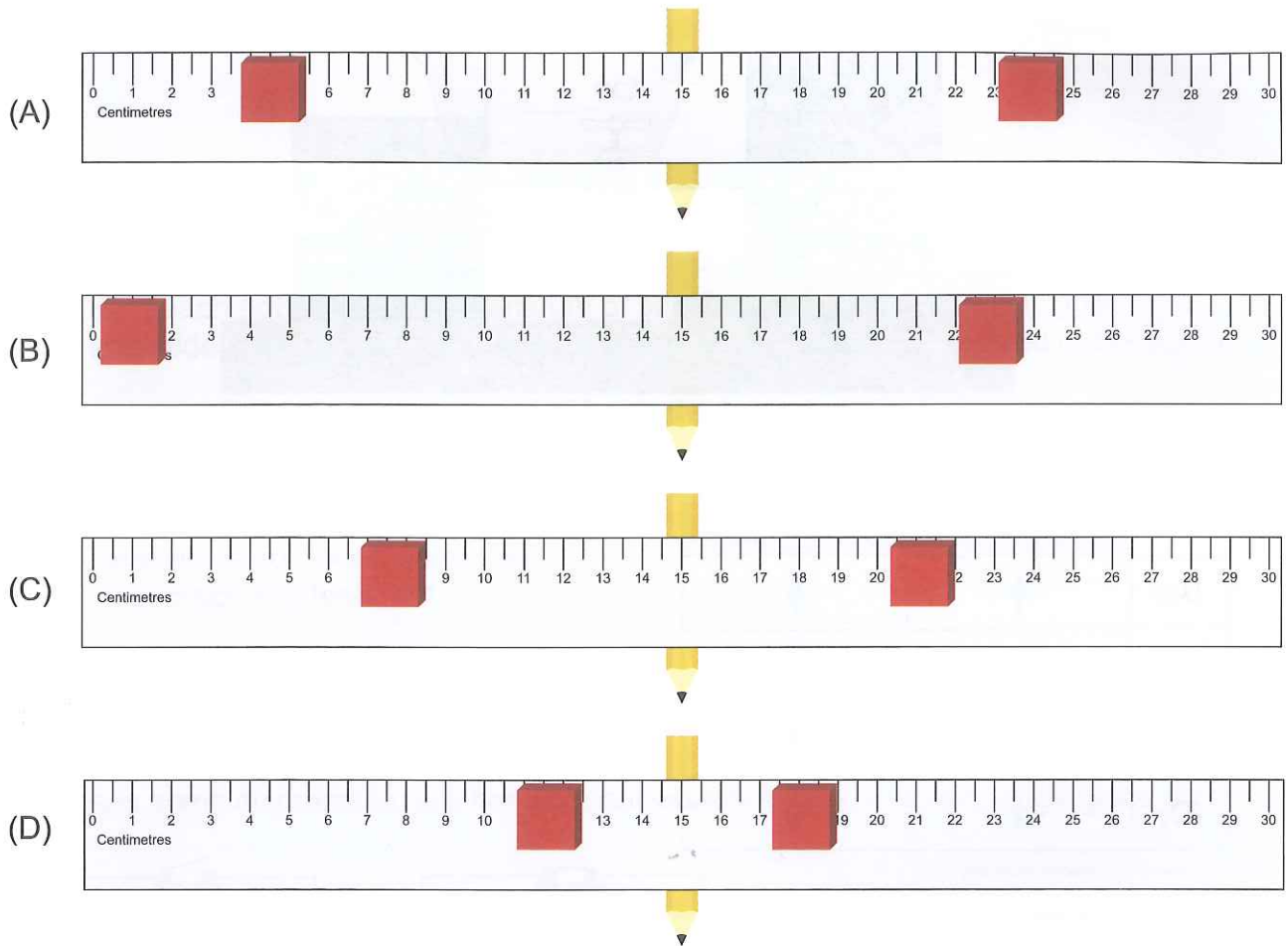
18. Maree wants to test how well different balls bounce by dropping them all from the same height. She will drop the balls so they all fall one metre.

Which diagram shows a ball at the correct height to begin her experiment?



19. Joe has two blocks that are the same size and have the same mass. He wants to place the two blocks on the ruler, so that the ruler balances on the pencil.

Where should he place the two blocks so that the ruler balances?



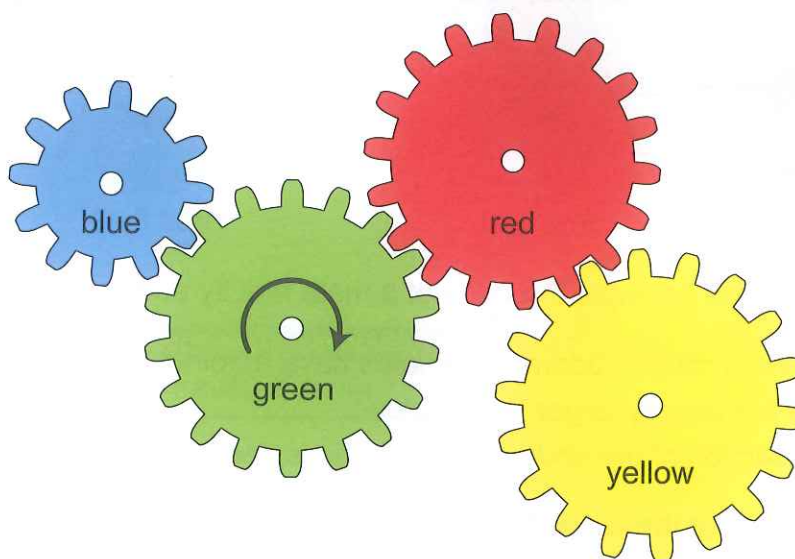
20. On 1 June, Jacob bought four brands of milk and placed them in his fridge. He tested how long it took each brand of milk to turn sour from its use-by date.

Brand	Use-by date	Date milk turned sour
Cow Valley	3 June	8 June
Milkman's Own	11 June	19 June
Cow Juice	8 June	18 June
Milko Farm	9 June	18 June

Which brand of milk took the longest time to turn sour from its use-by date?

- (A) Cow Valley
(B) Milkman's Own
(C) Cow Juice
(D) Milko Farm

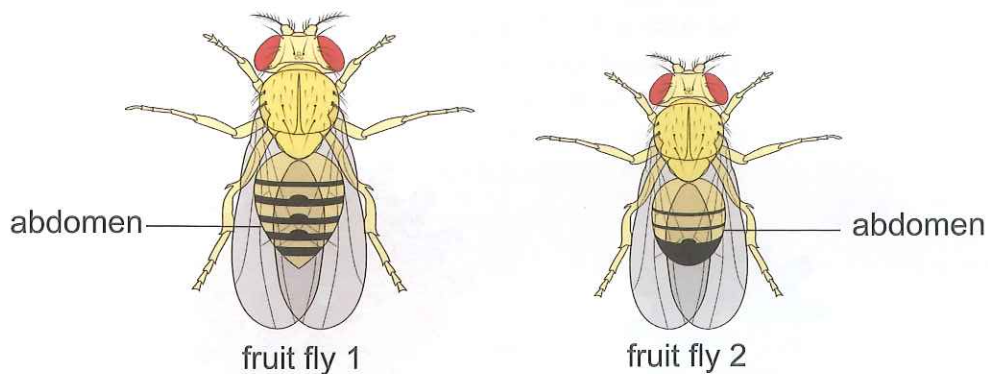
The green gear is turned in the direction shown by the arrow.



- | | blue gear | red gear |
|-----|---|---|
| (A) |  |  |
| (B) |  |  |
| (C) |  |  |
| (D) |  |  |

- (A) red (B) green
(C) blue (D) yellow

23. This picture shows two fruit flies.



There are differences between the body of a male fruit fly and that of a female fruit fly.

- Males have a round abdomen, females have a pointed abdomen.
- Females are usually larger than males.
- The last section of the abdomen in a male is darker than that of a female.

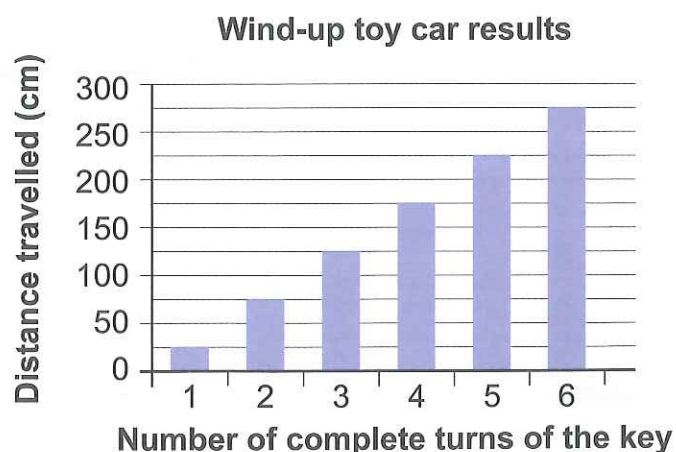
Which sentence about these two fruit flies is correct?

- (A) They are both male.
- (B) They are both female.
- (C) Fruit fly 1 is a male and fruit fly 2 is a female.
- (D) Fruit fly 1 is a female and fruit fly 2 is a male.

24. Wind-up toy cars move forwards when you wind them up with a key and let them go.

Frank carried out an investigation to find out if the number of times he turns the key affects the distance the car travels.

The graph shows Frank's results.



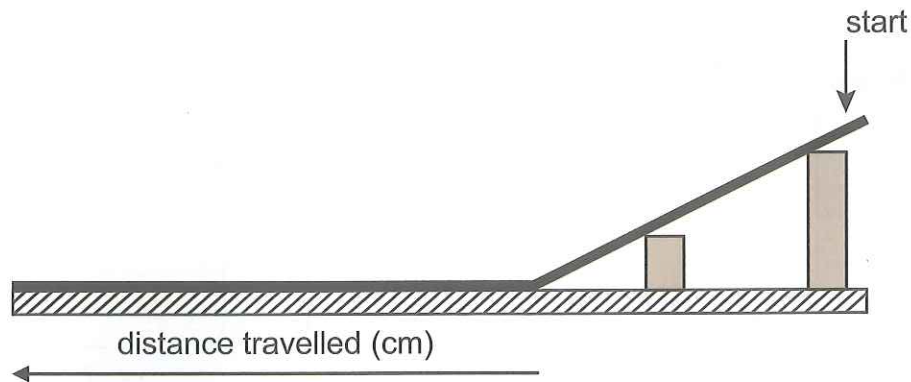
What is the least number of complete turns of the key needed to make the car travel 175 cm?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

For questions 25 and 26 use the information below.

Lisa has four different-sized marbles. She wants to see if the size of the marble is the only thing that affects how far it travels when it rolls down a ramp.

The diagram shows how she set up her experiment.



The table gives the size and mass of her marbles.

	 marble 1	 marble 2	 marble 3	 marble 4
size of marble (cm)	$\frac{1}{2}$	1	$1\frac{1}{2}$	2
mass of marble (gram)	5	12	12	25

25. Which two marbles should Lisa use to check whether size affects how far it travels?

- (A) 1 and 2
- (B) 2 and 3
- (C) 3 and 4
- (D) 4 and 1

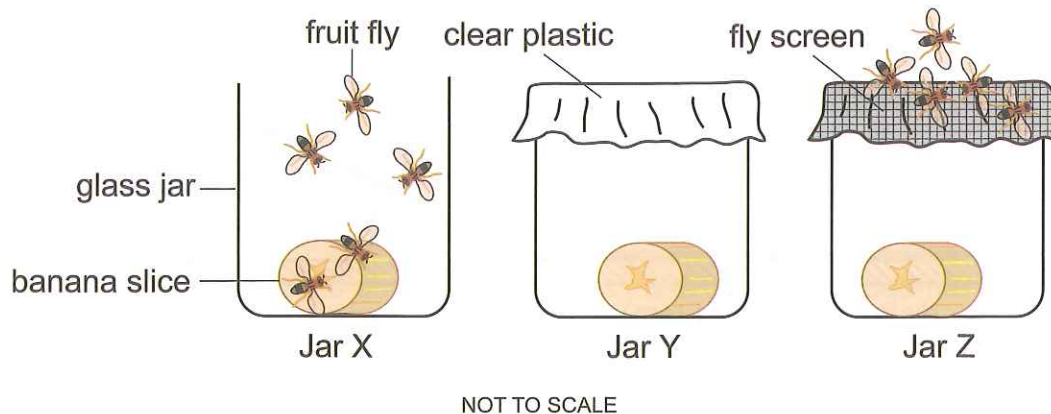
26. What can Lisa do to make the marble roll faster?

- (A) decrease the height of the ramp
- (B) increase the height of the ramp
- (C) make the distance travelled longer
- (D) make the distance travelled shorter

For questions 27 and 28 use the information below.

Nathan wanted to find out how fruit flies find bananas.

Nathan cut three pieces from one banana and placed them into three identical glass jars. He covered two of the jars with different materials. He placed all three jars together on a table outside. The diagrams show what Nathan observed a short time later.



27. Based on these observations, how do fruit flies find bananas?

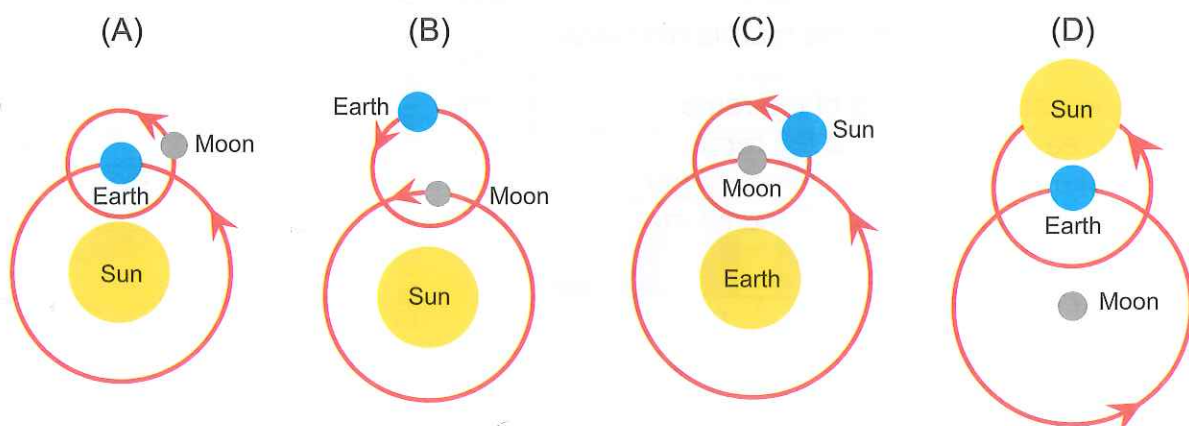
- | | |
|------------------------------|------------------------|
| (A) by sight | (B) by smell |
| (C) by either sight or smell | (D) by smell and taste |

28. Which condition did Nathan have to keep the same for each jar to make his investigation a fair test?

- | | |
|-------------------------------|----------------------------------|
| (A) the type of cover | (B) the type of fruit fly |
| (C) the number of fruit flies | (D) the size of the banana slice |

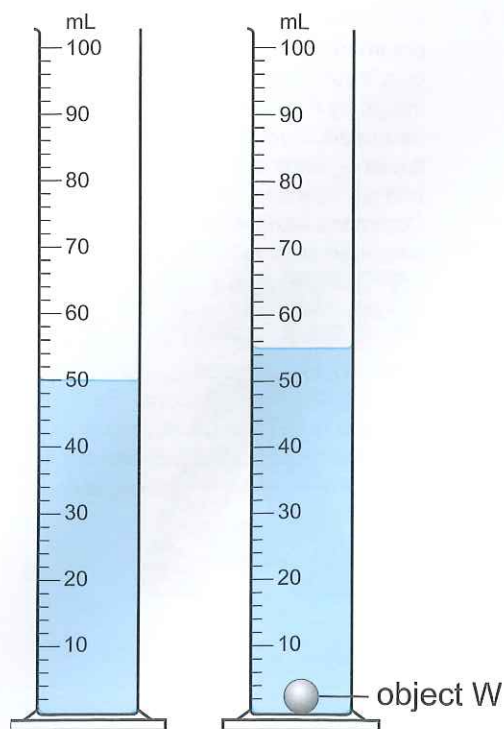
29. Earth moves around the Sun. As Earth moves around the Sun, the Moon moves around Earth.

Which diagram best represents the movement of Earth and the Moon?



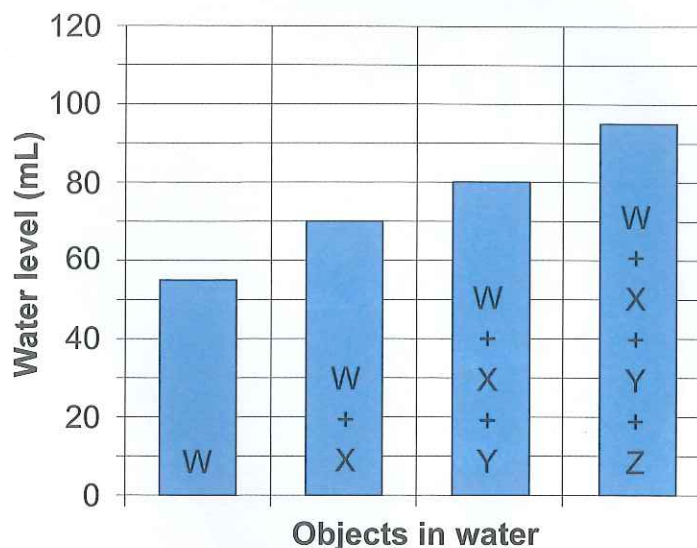
30. Ted has four objects: W, X, Y and Z.

Ted adds 50 millilitres (mL) of water to a measuring cylinder. He places object W into the water and records the new water level.



He then adds each object in turn and records the new water level.

The column graph shows his results.



Which two objects have the same volume?

(A) W and X

(B) X and Y

(C) X and Z

(D) Y and Z

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Sources

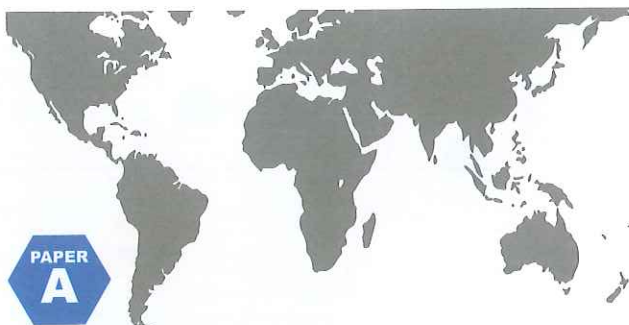
Question 1 Hummingbird image by Jônatas Cunha <<<http://commons.wikimedia.org/wiki/File:Beehummingbird.jpg>>> licensed under Creative Commons Attribution-ShareAlike 2.0 Generic licence <<<http://creativecommons.org/licenses/by-sa/2.0>>>.

Question 2 Whooper swan feet image by Aomorikuma <<http://commons.wikimedia.org/wiki/File:Whooper_Swan_feet.JPG>>. Little egret feet image by J.M.Garg <<[http://commons.wikimedia.org/wiki/File:Little_Egret_\(Egretta_garzetta\)_feet_in_AP_W2_IMG_3625.jpg](http://commons.wikimedia.org/wiki/File:Little_Egret_(Egretta_garzetta)_feet_in_AP_W2_IMG_3625.jpg)>>. Steller's sea eagle feet image by 4028mdk09 <<http://commons.wikimedia.org/wiki/File:Riesensee_adler_Fang.JPG>>. American herring gull feet image by Fcb981 <<http://commons.wikimedia.org/wiki/File:Larus_smithsonianus_feet1.jpg>>. Images licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported licence <<<http://creativecommons.org/licenses/by-sa/3.0>>>.

Question 4 Red hibiscus image by KENPEI <<http://commons.wikimedia.org/wiki/File:Hibiscus_coccineus1.jpg>>. Great white trillium image by ForestWander <<http://commons.wikimedia.org/wiki/File:White-trillium-flower_-_Virginia_-_ForestWander.jpg>>. Images licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported licence <<<http://creativecommons.org/licenses/by-sa/3.0>>>. Western red lily image by Wing-Chi Poon <<http://commons.wikimedia.org/wiki/File:Wild_Lily_Blooming_along_Dog_Lake_Trail_at_Kootenay_National_Park.jpg>> licensed under the Creative Commons Attribution-ShareAlike 2.5 Generic licence <<<http://creativecommons.org/licenses/by-sa/2.5>>>.

The following year levels should sit THIS Paper:

Australia	Year 3
Brunei	Primary 3
Indonesia	Year 4
Malaysia	Standard 3
New Zealand	Year 4
Pacific Region	Year 4
Singapore	Primary 2
South Africa	Grade 3



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HOW TO FILL OUT THIS SHEET:



- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.
- Rub out all mistakes completely.

EXAMPLE 1: Debbie Bach

FIRST NAME

D	E	B	B	I	E
A	A	A	A	A	A
B	B			B	B
C	C	C	C	C	
D	D	D			
E		E			

LAST NAME

B	A	C	H
A		A	A
	B	B	B
C	C		
D			
E			

EXAMPLE 2: Chan Ai Beng

FIRST NAME

C	H	A	N
O	A	B	E
A	A	A	A
B	B	B	B
C	C	C	C
D	D	D	D
E	E	E	E

LAST NAME

A	I	B	E	N	G
O	A	A	A	A	A
B	B	B	B	B	B
C	C	C	C	C	C
D	D	D	D	D	D
E	E	E	E	E	E

EXAMPLE 3: Jamal bin Abas

FIRST NAME

J	A	M	A	L		B	I	N
A		A		A		A	A	C
B	B	B	B	B		B		
C	C	C	C	C		C		
D	D	D	D	D		D		
E	E	E	E	E		E		

LAST NAME

A	B	A	S
A		A	C
B		B	
C		C	
D		D	
E		E	

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / / Postcode:

DATE OF BIRTH

Day		Month		Year	
0	0	0	0	0	0
1	1	1	1	1	1
2	2		2	2	2
3	3		3	3	3
4			4	4	4
5			5	5	5
6			6	6	6
7			7	7	7
8			8	8	8
9			9	9	9

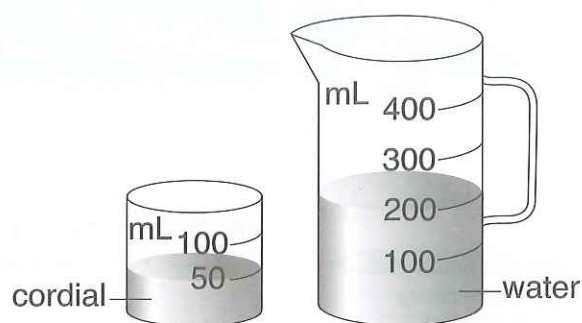
CLASS
(optional)

	(A)	(K)
	(B)	(L)
	(C)	(M)
	(D)	(N)
	(E)	(O)
	(F)	(P)
	(G)	(Q)
	(H)	(R)
	(I)	(S)
	(J)	(T)

TO ANSWER THE QUESTIONS

Example: Ari added cordial to water to make a jug of drink.
What will be the volume of the drink in the jug?

- (A) 50 mL
- (B) 150 mL
- (C) 200 mL
- (D) 250 mL



The answer is 250 mL, so you would fill in the oval **(D)**, as shown.

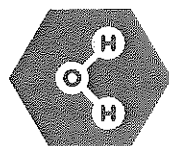
(A) (B) (C) **(D)**



START

- 1 (A) (B) (C) (D)
- 2 (A) (B) (C) (D)
- 3 (A) (B) (C) (D)
- 4 (A) (B) (C) (D)
- 5 (A) (B) (C) (D)
- 6 (A) (B) (C) (D)
- 7 (A) (B) (C) (D)
- 8 (A) (B) (C) (D)
- 9 (A) (B) (C) (D)
- 10 (A) (B) (C) (D)
- 11 (A) (B) (C) (D)
- 12 (A) (B) (C) (D)
- 13 (A) (B) (C) (D)
- 14 (A) (B) (C) (D)
- 15 (A) (B) (C) (D)

- 16 (A) (B) (C) (D)
- 17 (A) (B) (C) (D)
- 18 (A) (B) (C) (D)
- 19 (A) (B) (C) (D)
- 20 (A) (B) (C) (D)
- 21 (A) (B) (C) (D)
- 22 (A) (B) (C) (D)
- 23 (A) (B) (C) (D)
- 24 (A) (B) (C) (D)
- 25 (A) (B) (C) (D)
- 26 (A) (B) (C) (D)
- 27 (A) (B) (C) (D)
- 28 (A) (B) (C) (D)
- 29 (A) (B) (C) (D)
- 30 (A) (B) (C) (D)



S

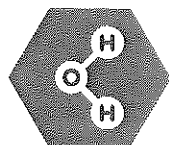
2015 Science Answer Keys

ICAS

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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	C	D	B	D	C	A	D	C	D
2	D	D	B	A	A	B	B	A	D
3	B	D	C	C	D	A	B	B	C
4	D	C	A	C	A	A	B	D	B
5	B	B	D	D	B	B	C	C	B
6	D	B	C	B	D	B	C	A	C
7	C	B	C	D	B	D	A	A	B
8	B	A	C	C	D	B	D	D	B
9	C	C	C	C	B	D	C	C	B
10	C	C	A	D	C	C	C	A	A
11	C	A	D	B	B	D	A	D	D
12	B	C	C	B	B	D	C	A	D
13	A	D	D	B	D	A	B	A	D
14	D	B	D	A	A	D	D	D	C
15	A	A	B	D	D	C	A	D	A
16	A	C	A	B	A	A	D	C	C
17	C	A	C	C	C	A	A	A	A
18	A	C	B	B	B	C	C	A	C
19	D	D	C	B	A	B	D	B	A
20	C	A	C	A	A	B	A	B	B
21	A	B	D	D	A	C	D	D	D
22	C	D	B	A	A	D	B	D	B
23	D	B	A	D	B	C	C	B	A
24	B	B	B	D	D	C	A	C	B
25	B	A	B	A	D	B	C	C	B
26	B	C	A	A	C	B	B	B	D
27	B	D	A	C	C	C	C	D	A
28	D	A	A	D	B	B	D	B	D
29	A	A	B	A	B	C	B	A	A
30	C	D	C	D	B	C	B	C	C

(Please turn over)



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2015 Science Answer Keys

ICAS

International Competitions
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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			A	A	C	B	D	B	D
32			C	D	C	C	B	A	C
33			C	A	A	D	D	C	A
34			D	B	B	D	A	B	D
35			D	D	C	B	D	C	A
36			B	B	C	C	B	B	C
37			D	C	B	C	C	A	B
38			B	D	C	D	A	A	C
39			D	C	C	A	A	A	C
40			D	C	C	A	B	D	B
41					B	D	D	B	B
42					D	A	B	B	A
43					D	A	A	C	B
44					D	A	B	B	C
45					C	C	A	D	D

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